SWEATERCO

GROUP 11 / TRACK 2 / SPRINT 2 REVIEW

StitchFlow

The Quick-Response Subcontracting Desk

Say the order. StitchFlow finds the right workshop.

PRIMARY MINIMISE LATENESS

SECONDARY MINIMISE DEFECTS

Chen Zepeng (A0156784N)
Present (in person)

Duong Thanh Nhan (A0296326W)
Absent: business trip to China

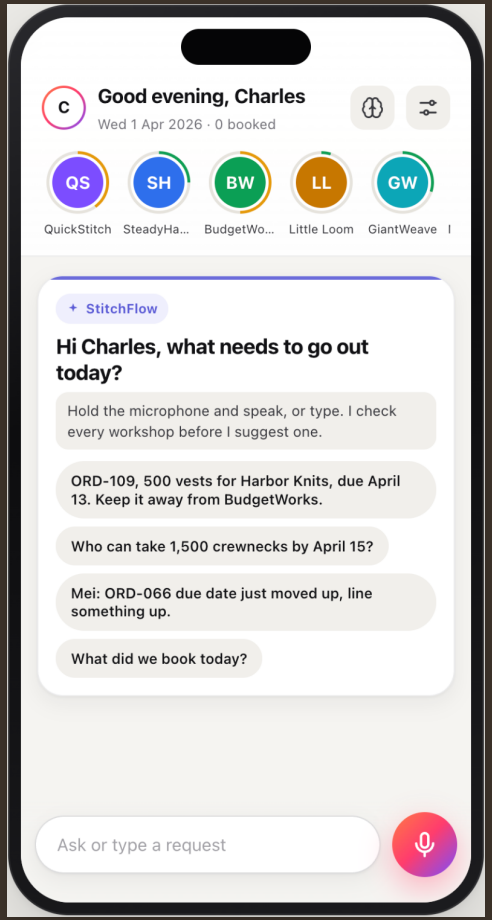
Joshua Wesley Gloor (A0296944J)
Present (in person)

Sridhar Niyanthri Rajeswari (A0351151U)
Present (in person)

Sundara Rajan Saadhvika (A0352594W)
Present (in person)

Tang Zheling (A0352177B)
Present (in person)

GITHUB github.com/JoshuaGloor/dss5105-capstone-project



DSS5105 CAPSTONE / 30 OCTOBER 2026

The dispatcher decides on the factory floor, in seconds

A REAL REQUEST FROM THE INBOX

"ORD-109, 500 vests for Harbor Knits, due Apr 13. ...and keep it away from BudgetWorks, the last batch came back rough."

Mei, 09:53, request R25

THE USER

The production dispatcher at SweaterCo

On the floor with a phone. Most requests are messy: missing dates, vague orders, banned workshops.

1. A safe workshop, its date and why, in seconds
2. The constraint kept, and who asked recorded
3. One deliberate step to commit; a record after

WHAT CHANGED SINCE SPRINT 1

68%
fewer late days than earliest finish, which tied us in Sprint 1 (1.42% late vs 1.67%)

4.90%
defective pieces with the defects setting, vs 5.28% for biggest capacity, which won in Sprint 1

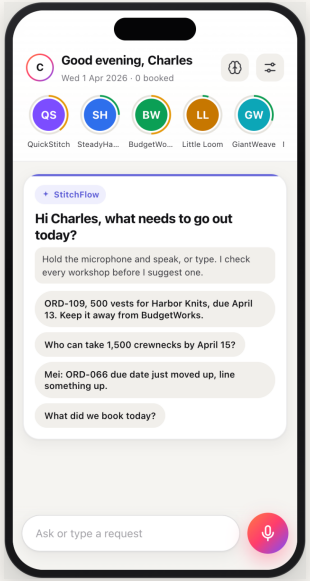
19.2%
largest workshop share with the fairness setting, vs 22.6% for random; not tuned in Sprint 1

NEW **Voice on the phone: answers read aloud, and one slide to book**

NEW **Memory: confirmed constraints come back, with who said them**

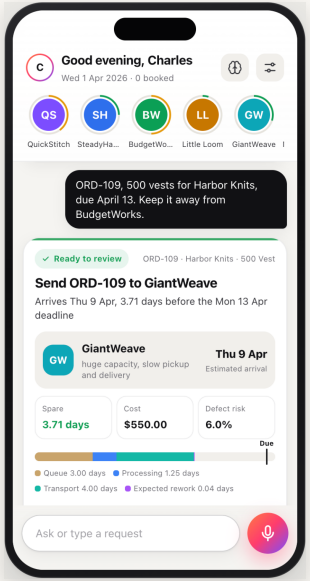
Course simulator, mean of 10 normal runs (seeds 5105-5114); presets chosen on development seeds first. docs/OBJECTIVE_STUDY.md

Say it, check it, slide to book, and it remembers



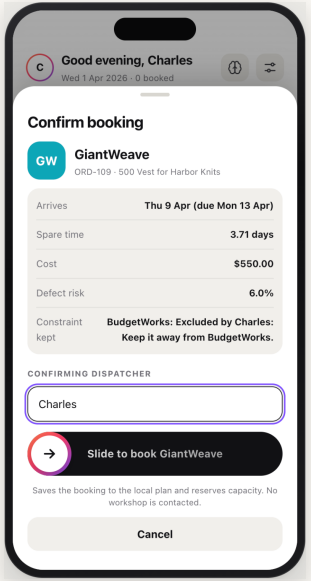
1 Ask

Hold the mic: order, date and any constraint, in plain words.



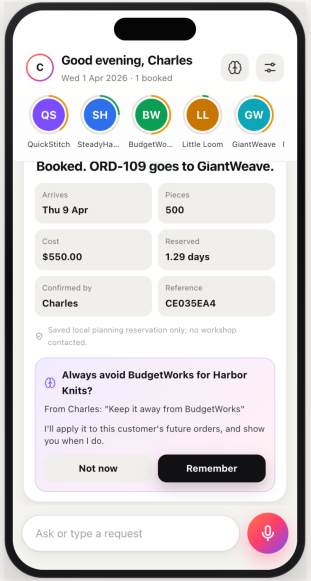
2 Answer

Best workshop, arrival and spare days, and why. Read aloud.



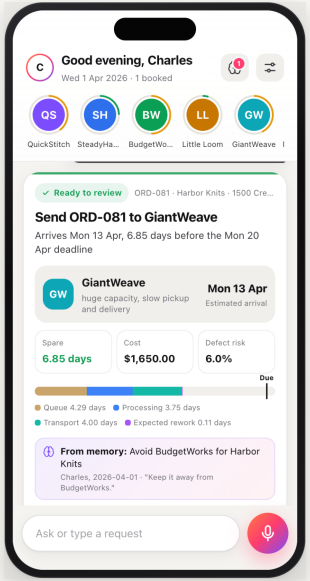
3 Confirm

"Book it" opens one sheet. One slide books it, with a name on it.



4 Remember

It offers to keep the constraint for this customer.

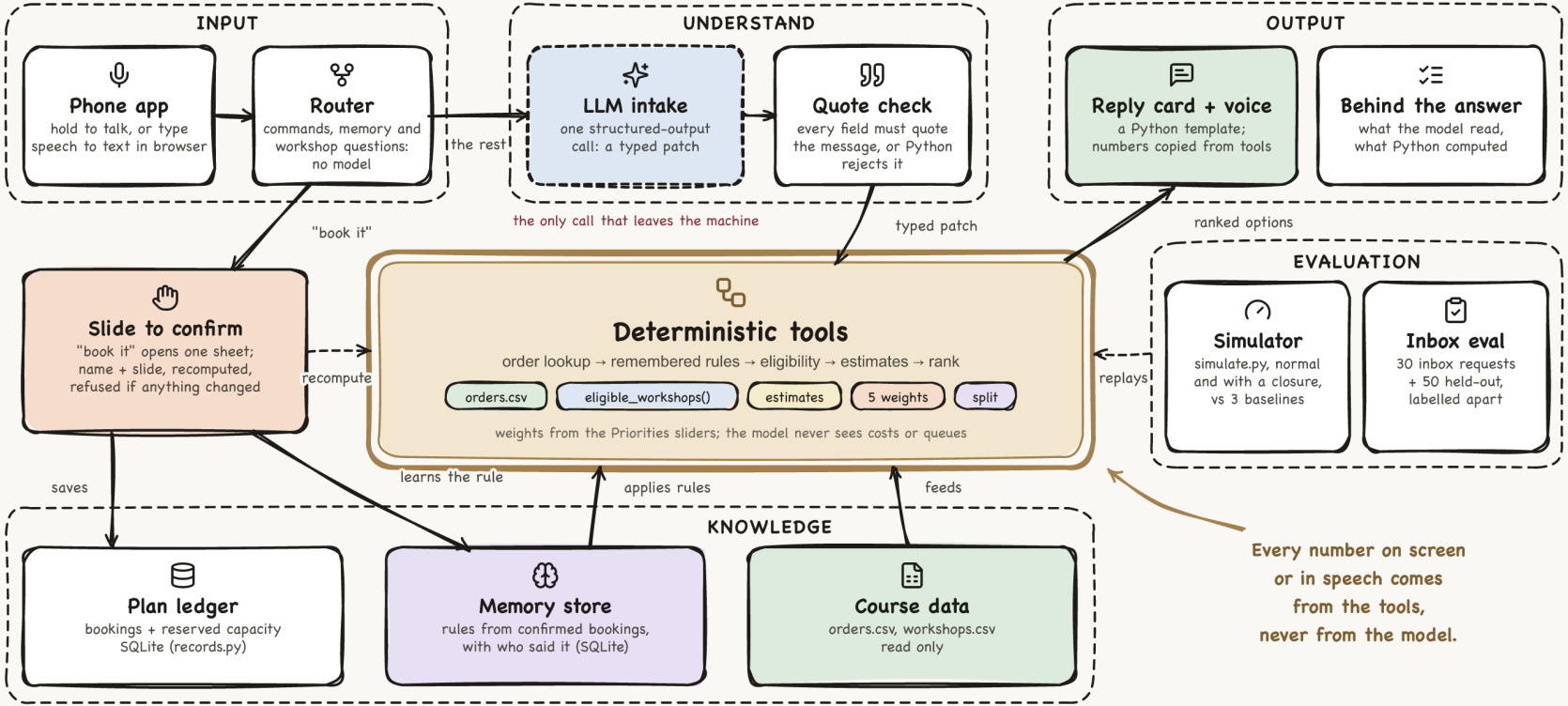


5 Reuse

Next order: the rule is applied, and the card says so.

Live screenshots of the running app (local plan, model: the one set in .env). Voice proposes; only the slide commits.

The model reads, Python computes, a person commits



Each choice, against the alternative we considered

MODEL

One hosted model, named in .env

over a fine-tuned or local model

It only extracts quoted fields, so no training is needed; swapping it is one line.

RETRIEVAL

Exact lookup by order ID

over embedding retrieval (RAG)

Orders have keys; an exact match cannot fetch the wrong order.

DATA STORE

SQLite for records and memory

over a vector store

A rule must apply exactly, with who said it, auditable in one local file.

TOOLS

The course's eligible_workshops()

over our own eligibility rule

Our checks and the course simulator can never disagree.

ORCHESTRATION

One fixed workflow

over a ReAct agent loop

Every request needs the same checks in order: testable, cheaper, no tool-choice errors.

REPLIES

Python templates

over model-written replies

No model arithmetic: dates and costs are copied from tool outputs.

SPEECH

Browser speech recognition

over server speech-to-text

Free, live words while speaking, no extra key; typing always works.

COMMIT

Slide to confirm

over booking on a spoken "yes"

A misheard "yes" on a noisy floor cannot send work out.

Highlighted: the four choices the brief asks us to justify. The model call is the only thing that leaves the machine; it never sees costs, queues or defect rates.

Every feature, its owner and its status

7/11

features done this sprint

Each card names its owner. Code and pull requests are linked below.

DONE 7

- **Voice assistant: speech in and out, slide to book**
  Chen Zepeng
- **Long-term memory of confirmed rules, log, profile**
  Chen Zepeng
- **Four-objective scorer, presets, objective lab**
  Chen Zepeng
- **Capacity questions and workshop checks from chat**
  Chen Zepeng
- **Chat-first desk redesign**
  Duong Thanh Nhan
- **AWS deployment behind HTTPS (demo site)**
  Joshua Wesley Gloor
- **Evaluation sets: 205 requests, 140 train / 65 test**
  Saadhvika and Niyanthri

IN PROGRESS 2

- **Clarification reliability fixes and 247-assertion benchmark**
  Tang Zheling
- **Independent labels for the 30 inbox requests**
  Saadhvika and Niyanthri

PLANNED 2

- **Re-run the 65 test requests on the current intake (v3)**
  Tang Zheling
- **Usability test: two people acting as dispatchers, four tasks**
  Joshua Wesley Gloor

Code: github.com/JoshuaGloor/dss5105-capstone-project. PRs: #15 and #18 this sprint, #16 and #17 evaluation sets, #14 clarification reliability.

What we measure, against what, and the hard cases

1.42%
late batches (our objective), vs 1.67% for earliest finish

0.057
mean late days, vs 0.178 for earliest finish

4.90%
defective pieces, vs 5.28% for biggest capacity

96.9%
of 65 held-out requests passed (63/65); train 95.7%

WHAT WE MEASURE, AND WHY

Late batches and late days (our objective), defective pieces, largest share and the change under a closure: the course simulator's own metrics. Language: the right behaviour per request, every number traced to a tool.

HOW THE EVALUATION SET IS BUILT

30 inbox requests by kind, plus 205 generated by Saadhvika and Niyanthri: 140 to tune on, 65 held out with new wording. Labels come from how each request was built, never from a model.

BASELINE AND TRACK MINIMUM

Random, biggest capacity and cheapest, plus earliest finish. Minimum: beat all three baselines on late batches and late days, and handle all 30 inbox requests by kind.

HARD CASES AND HOW WE HANDLE THEM

1 **Impossible date**
Earliest date, days late and a two-shop split; nothing is booked

2 **Workshop that cannot take it**
Refused with the reason, plus the best permitted alternative

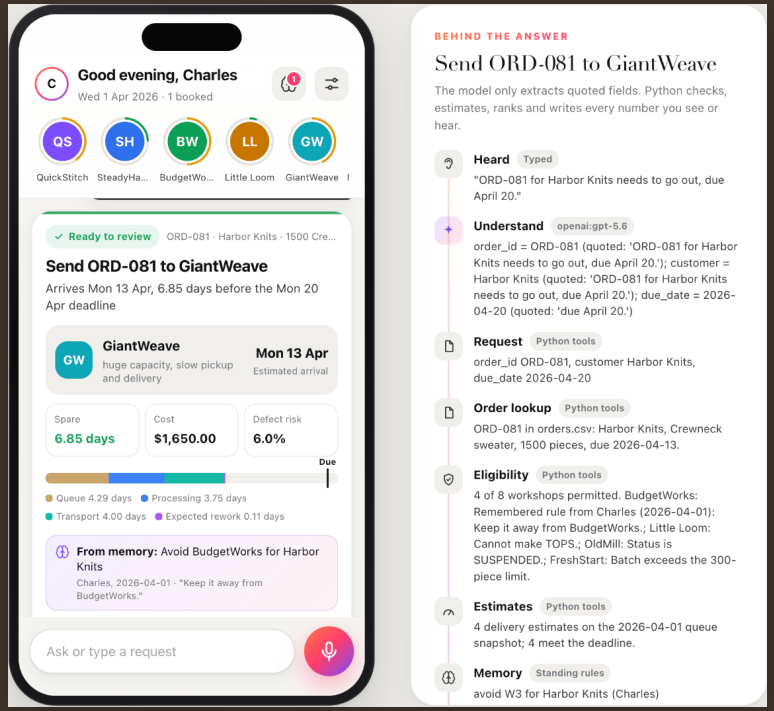
3 **Misheard order on the floor**
Order, customer and pieces read back from orders.csv; a slide commits

Allocation: course seeds 5105-5114, presets chosen on seeds 6000-6029 (OBJECTIVE_STUDY.md). Language: one gpt-5.6 run; a rubric pass rate, not model accuracy.

Charles, as the dispatcher

Presented by Chen Zepeng (A0156784N), who goes by Charles

1. Says a real inbox request out loud
2. Hears the best workshop and why
3. Says "book it" and slides to confirm
4. Lets it remember the constraint
5. A new order: the rule is applied
6. Hard case: a workshop that cannot take it, and an impossible date
7. Behind the answer: what the model read, what Python computed



S T I T C H F L O W

Thank you.

Questions are welcome.

GITHUB github.com/JoshuaGloor/dss5105-capstone-project

SAY THE ORDER / PYTHON CHECKS / A PERSON COMMITS

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